PROJECT DOCUMENTATION

EV-Market-Analysis

Electric Vehicle Market Performance & Insights

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Course: B.Tech

College / University: Medicaps University

Visualization Tool: Tableau

Prepared for academic project submission

1. Introduction

Electric vehicles (EVs) are becoming an important part of the transportation industry as manufacturers and governments focus on cleaner and more efficient mobility. Market data can help users understand sales patterns, vehicle-category performance, regional distribution, and battery technology trends.

This project, EV-Market-Analysis, uses Tableau to present EV market data through charts and dashboard visualisations. The project overview highlights the importance of analysing sales, vehicle categories, battery technologies, and market regions.

2. Problem Statement

EV market data can be difficult to interpret when it is presented only in tables. A visual dashboard makes it easier to compare categories, observe differences between regions, and understand the distribution of battery types. The project aims to transform EV market data into clear visualisations that support quick exploration and interpretation.

3. Objectives

• To explore electric-vehicle market data using Tableau.
• To compare annual sales units across vehicle categories.
• To visualise the distribution of market sales by region.
• To show the share of different battery types.
• To present the findings in a single, understandable dashboard.

4. Tools and Technologies

Tableau: used to create charts, visualisations, and the dashboard.

EV market dataset: the source data used for sales, category, region, and battery-type analysis. The exact source and file details should be added after verification.

GitHub: used to store project files, screenshots, README, and documentation, if included in the submission.

5. Data Collection and Preparation

The project uses EV market data containing fields represented in the dashboard, including vehicle category, annual sales units, market region, and battery type.

Before final submission, verify the dataset source, record count, time period, and column definitions. Check for missing values, inconsistent category names, duplicate records, and incorrect data types before building the visualisations.

Note: Dataset source and cleaning operations were not provided, so they should be completed with the actual project details rather than assumed.

6. Methodology

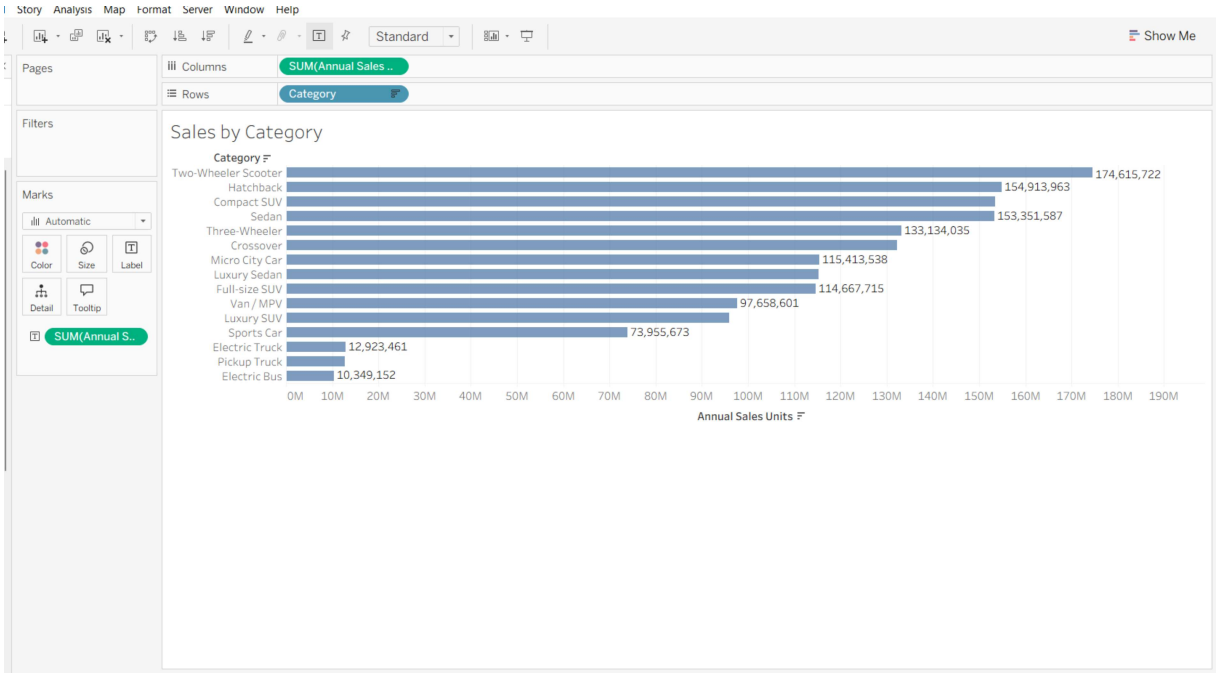
Step 1 — Obtain and open the EV market dataset.
Step 2 — Connect the dataset to Tableau.
Step 3 — Check field names, data types, missing values, and duplicate records.
Step 4 — Build a bar chart using vehicle Category and Annual Sales Units.
Step 5 — Build a donut chart to compare sales by Market Region.
Step 6 — Build a pie chart to show the Battery Type distribution.
Step 7 — Arrange the visualisations on a dashboard and add titles, labels, and legends.
Step 8 — Review the dashboard for readability and verify values against the source data.
Step 9 — Save the workbook and upload the required files and documentation.

7. Dashboard and Visualisations

The following figures are the Tableau visualisations supplied for this project. They are placed directly in Section 7 for easy reference.

Figure 1. Sales by Category

Horizontal bar chart comparing annual sales units across vehicle categories.



7. Dashboard and Visualisations (continued)

Figure 2. Sales by Market Region

Donut chart comparing the displayed sales values across market regions.

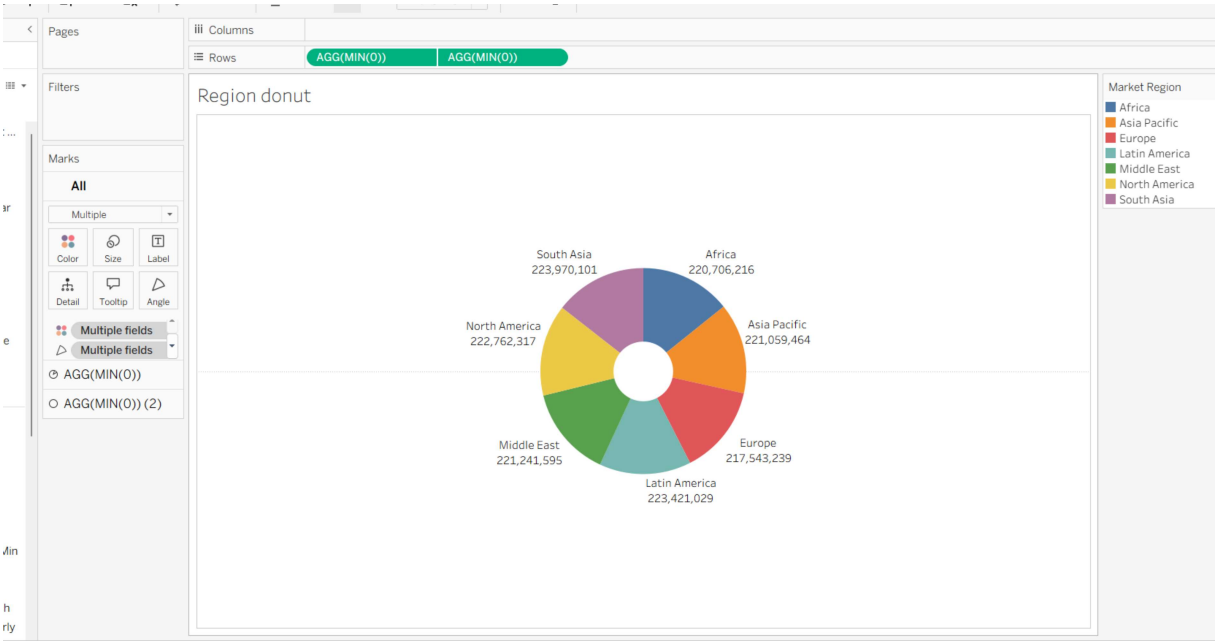
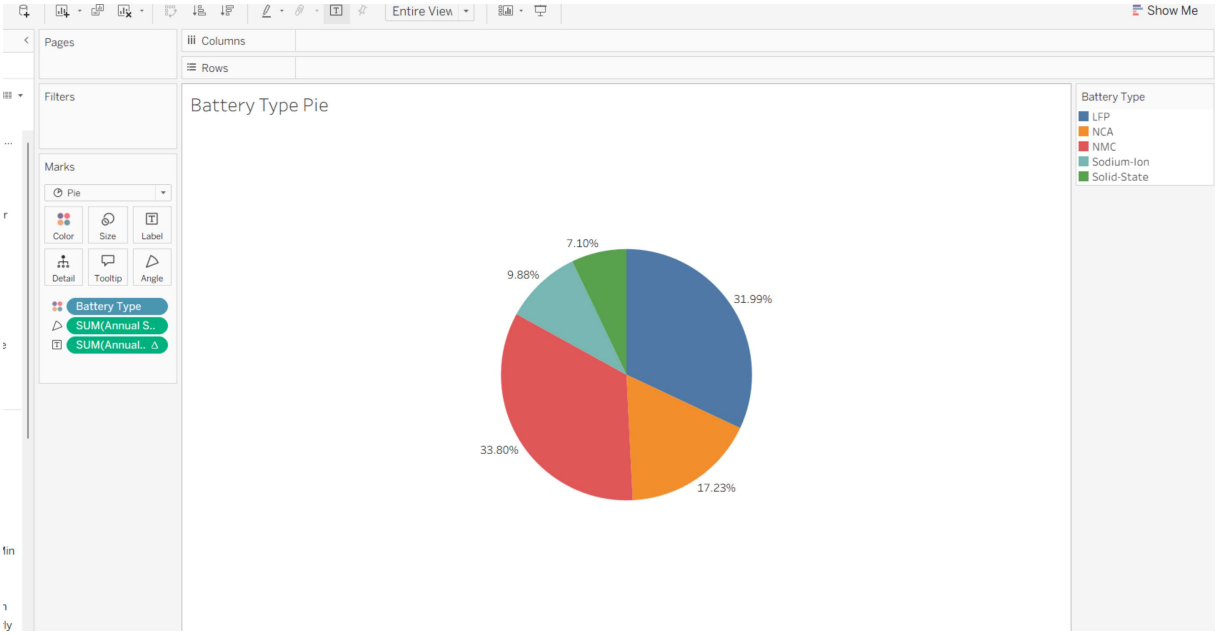


Figure 3. Battery Type Distribution

Pie chart showing the displayed share of each battery type.



7. Dashboard and Visualisations (continued)

Figure 4. Sales Trend by Model Year

Line chart showing annual sales units across model years.

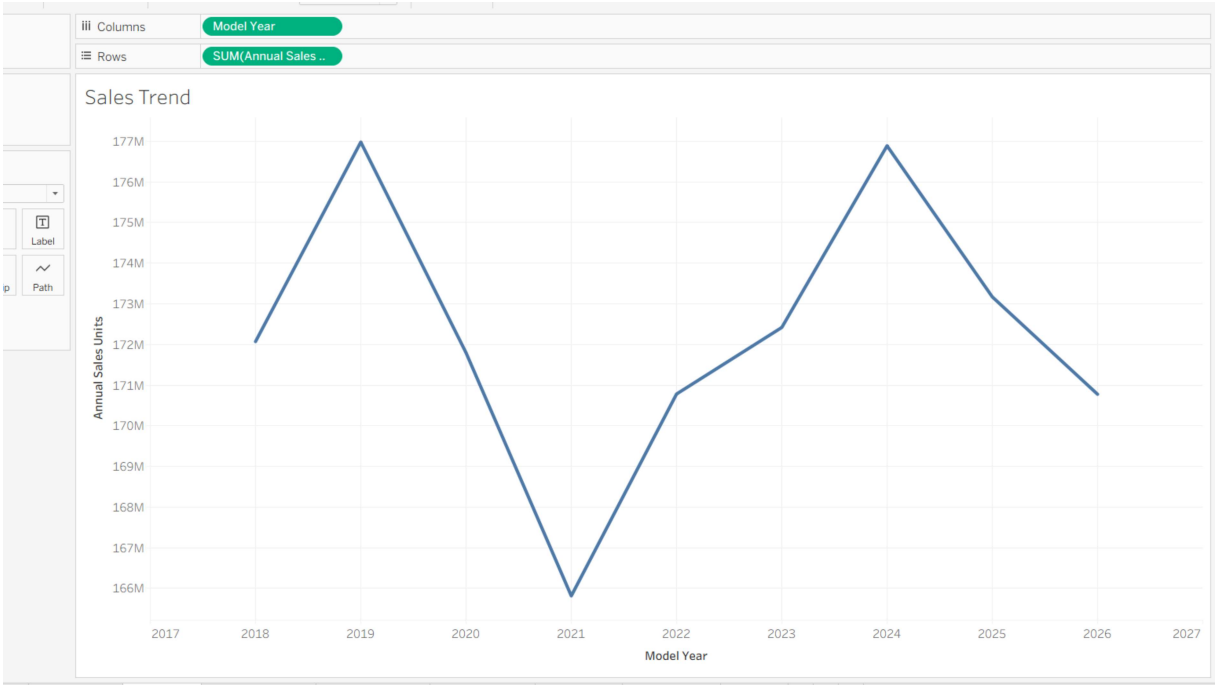
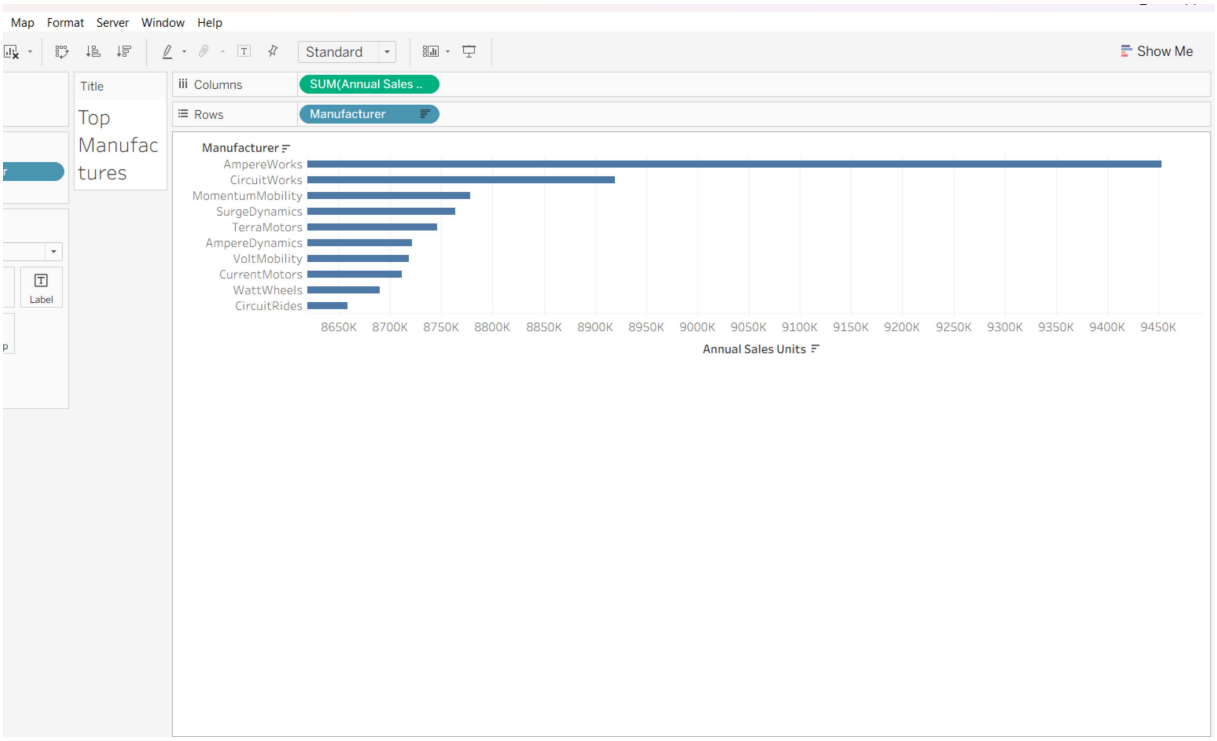


Figure 5. Top Manufacturers

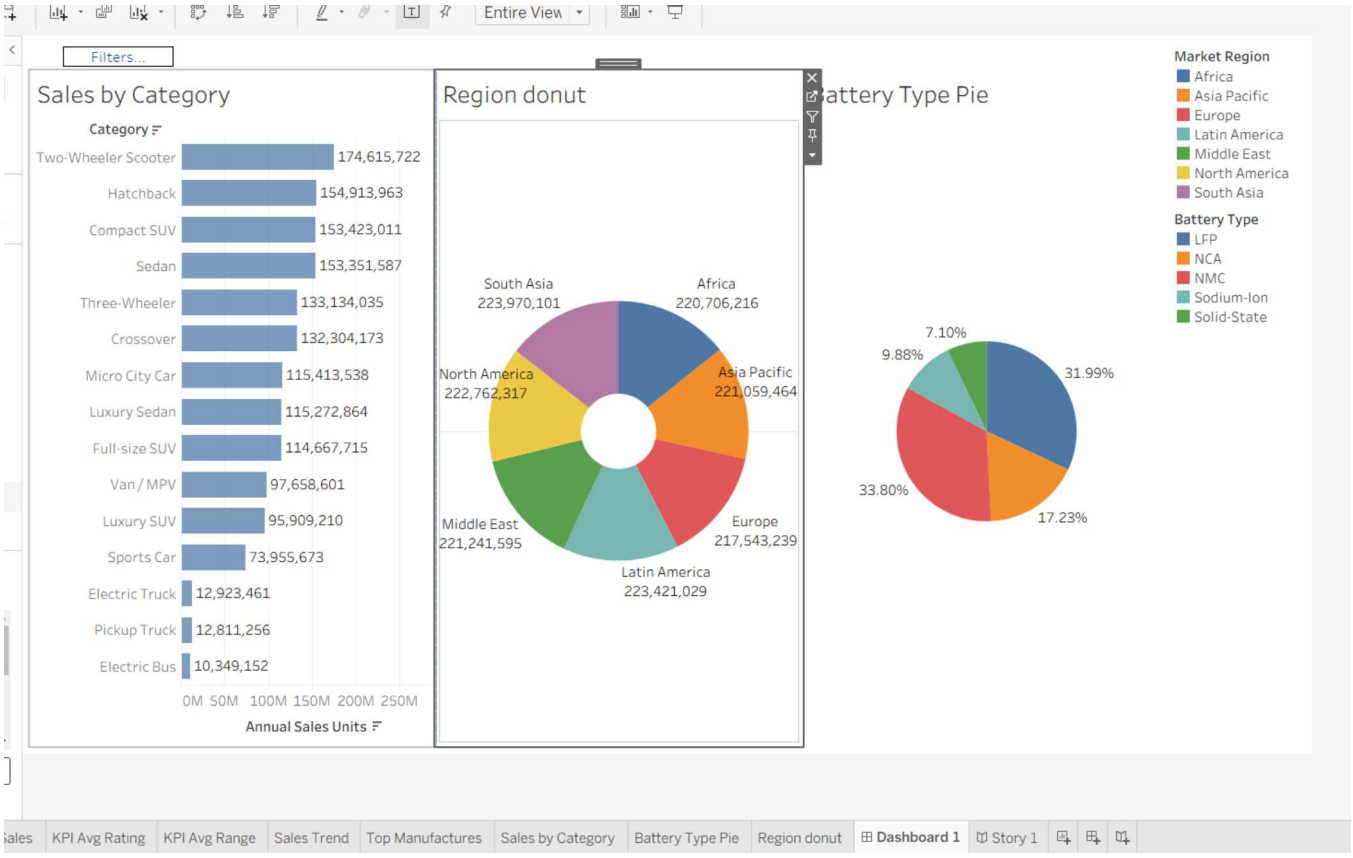
Horizontal bar chart comparing displayed annual sales units by manufacturer.



7. Dashboard and Visualisations (continued)

Figure 6. Complete Tableau Dashboard

Dashboard view combining Sales by Category, Region donut, and Battery Type pie charts.



Source: Tableau dashboard screenshot supplied for the project.

8. Key Observations

Based on the provided dashboard screenshots, Two-Wheeler Scooter is the highest displayed sales category, while Electric Bus is the lowest displayed category.

The regional values displayed in the donut chart are relatively close to one another, with South Asia the highest and Europe the lowest among the listed values.

The battery-type pie chart shows NMC as the largest displayed share (33.80%), followed by LFP (31.99%). Solid-State has the smallest displayed share (7.10%).

These are observations from the supplied screenshots, not conclusions about the entire EV industry. They should be validated against the source data and chart configuration.

9. Expected Outcome

The expected outcome is an interactive Tableau dashboard that summarises EV market information using category sales, regional comparison, and battery-type distribution. It should make the data easier to compare and support further analysis.

10. Limitations and Future Scope

Limitations: the dataset source, time period, and data-cleaning details have not yet been supplied. The dashboard values and aggregation settings should be verified in Tableau.

Future scope: add filters for model year and region, include a sales trend over time, compare manufacturers, add average rating or driving-range KPIs if available, and publish a shareable Tableau demo link.

11. Conclusion

The EV-Market-Analysis project demonstrates how Tableau can be used to convert electric-vehicle market data into visual insights. The dashboard compares annual sales across vehicle categories, displays regional values, and summarises battery-type shares. Verifying the source data and adding the final repository and demo links will make the submission more complete.